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Functional Requirements

EGEN5208W Databases for Soft Engineers

Course: EGEN5208W Databases for Soft Engineers

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Required Operations

You must implement exactly eight (8) application operations.

Each operation:

- Must execute one or more SQL statements
- Must interact with actual database tables
- Must handle both success and failure cases
- Must be executable independently of other operations

Implementing more than 8 operations will not earn additional marks.

Role	Operations	Count
Member	User Registration, Profile Management, Health History, Dashboard	4
Trainer	Set Availability, Schedule View	2
Administrative Staff	Room Booking, Equipment Maintenance	2
Total		8

Member Functions (4 Operations)

1. User Registration

Create a new member account.

The system must:

- Accept the following inputs:
 - Full name
 - Email address (must be unique)
 - Date of birth
 - Gender
 - Phone number
- Automatically store the registration timestamp
- Reject duplicate email registrations with a clear error message

Required SQL behavior: `INSERT` to create the member record

2. Profile Management

Allow a logged-in member to manage their own profile and health data.

The system must allow:

- Updating basic profile information (e.g., name, phone)
- Creating or updating fitness goals (e.g., target weight)

- Inserting new health metric records

Health metrics:

- Must be stored as append-only records
- Must include:
 - Metric type (e.g., weight, heart rate)
 - Value
 - Timestamp
- Existing health records must never be overwritten

Required SQL behavior: `UPDATE` for profile data, `INSERT` for goals, `INSERT` for health metrics

3. Health History

Display the complete health metric history for the logged-in member.

The system must:

- Retrieve all health metric records for the member
- Display:
 - Metric type
 - Value
 - Timestamp
- Sort results consistently by timestamp
- Display a message if no records exist

No modification operations are allowed here.

Required SQL behavior: `SELECT` , `WHERE` (filter by member), `ORDER BY`

4. Dashboard

Display a summary view for the logged-in member.

The dashboard must show:

- The most recent value for each health metric type
- All active fitness goals
- The total number of group classes attended
- A list of upcoming personal training sessions, including:
 - Date
 - Start time
 - End time
 - Trainer name

Charts, trends, predictions, or visual analytics are not required.

Required SQL behavior: `SELECT` with `JOIN` s, `COUNT` , subqueries or `ORDER BY ... LIMIT 1`

Trainer Functions (2 Operations)

5. Set Availability

Allow trainers to define when they are available.

Availability:

- Is represented as individual time slots
- Each slot includes:
 - Date
 - Start time

- End time
- Recurring availability is not required

The system must:

- Prevent overlapping availability slots for the same trainer
- Validate that end time is after start time
- Allow updating or deleting existing slots

Required SQL behavior: `SELECT` to detect overlap, `INSERT` , `UPDATE` , `DELETE`

6. Schedule View

Display the trainer's upcoming assignments.

The system must show:

- Personal training sessions:
 - Date, time, member name, room
- Group fitness classes:
 - Date, time, class name, room
- Only future sessions
- Results sorted by date and start time

Required SQL behavior: `SELECT` , `JOIN` , `WHERE` , `ORDER BY`

Administrative Staff Functions (2 Operations)

7. Room Booking

Allow administrators to assign rooms to sessions or classes.

The system must:

- Prevent assigning the same room to overlapping time periods
- Allow modifying an existing room assignment
- Reject conflicts with a clear error message

Room availability display:

- May be a simple list of rooms not already booked in a given time window

Required SQL behavior: `SELECT` to detect conflicts, `UPDATE` to assign or change rooms

8. Equipment Maintenance

Allow administrators to manage equipment records.

The system must support:

- Logging a maintenance issue
- Updating equipment status:
 - operational
 - under repair
 - out of service
- Viewing all equipment
- Filtering equipment requiring maintenance

Equipment must be associated with a room or location.

Required SQL behavior: `INSERT` , `UPDATE` , `SELECT` with `WHERE`

Implementation Notes

Each operation must be demonstrated with:

- One successful execution
- One failure case

All validation must be enforced through:

- SQL constraints, and/or
- Triggers, and/or
- Application-level checks

No hard-coded data is allowed. Errors must be handled gracefully.

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